

CELL AND MOLECULAR BIOLOGY LABORATORY

From Gene Mutation to Disease

Investigating How DNA Sequence Changes Affect Protein Products and Human Phenotypes

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Part I. Disease Background

Human Immunodeficiency Virus (HIV) is a disease that weakens the immune system by attacking CD4⁺ T cells, which help the body fight infections. To infect these cells, HIV first attaches to the CD4 receptor and then uses the CCR5 receptor as a co-receptor to enter the cell. As the virus continues to multiply, the number of CD4⁺ T cells decreases, making the body more vulnerable to infections and other diseases.

A well-known genetic variant of the CCR5 gene is the CCR5 $\Delta 32$ mutation, which is caused by the deletion of 32 nucleotides in the coding sequence. This mutation produces a shortened and nonfunctional CCR5 protein that cannot be properly expressed on the cell surface. Without a functional CCR5 receptor, most CCR5-tropic HIV-1 strains have difficulty entering CD4⁺ T cells. Because of this, people who inherit two copies of the $\Delta 32$ mutation have a high level of natural resistance to many HIV infections.

Part II. Research the Disease

A. Disease

15. What is the disease or phenotype?	HIV susceptibility is a phenotype describing how genetic and immune factors influence an individual's risk of acquiring HIV or progressing to AIDS.
14. What are its major clinical characteristics?	Its clinical characteristics include differences in HIV acquisition risk, viral load levels, CD4 ⁺ T-cell decline, and speed of progression to AIDS.
15. Which cells, tissues, or organs are mainly affected?	The main cells affected are CD4 ⁺ T lymphocytes, macrophages, and dendritic cells, with lymphoid tissues and mucosal sites being key targets.
16. What is its genetic basis?	The genetic basis involves variants in immune-related genes such as CCR5 (e.g., CCR5- $\Delta 32$ mutation), HLA alleles, and other genes like APOBEC3G and TRIM5 α that alter viral entry and immune defense.
17. What is its inheritance pattern, if	Its inheritance pattern can be autosomal

applicable?	recessive for CCR5-Δ32 resistance, but overall HIV susceptibility is polygenic and multifactorial, shaped by multiple genes and environmental influences.
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B. Gene and Normal Protein

18. What is the official gene symbol?	The official gene symbol is CCR5.
19. On which human chromosome is the gene located?	CCR5 is located on human chromosome 3 (3p21.31).
20. What does the gene normally encode?	It normally encodes the C-C chemokine receptor type 5 protein.
21. What is the normal biological function of the protein?	The protein functions as a cell surface receptor for chemokines, helping regulate immune cell trafficking and inflammatory responses.
22. Where in the cell is the protein normally found?	The CCR5 protein is normally found on the cell membrane of immune cells, such as T lymphocytes, macrophages, and dendritic cells.
23. In what biological pathway or cellular process does it participate?	It participates in immune signaling pathways, specifically in chemokine-mediated leukocyte migration, and is also exploited by HIV as a co-receptor for viral entry.

C. Documented Mutation

Gene	CCR5
Reference Transcript	NM_000579.3
Exact Variant Notation	NM_001394783.1:c.554_585del
Nucleotide Change	32-base pair deletion (Δ32)
Predicted Protein Change	NP_001381712.1:p.Ser185fs
Mutation Type	Frameshift deletion
ClinVar Accession when available	VCV000008184
Clinical Interpretation	Benign

Scientific references supporting the association	Samson et al., Nature (1996); Liu et al., Cell (1996); NCBI ClinVar CCR5 Δ32 record.1
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Hypothesis

If the documented CCR5 Δ32 mutation is introduced into the wild-type CCR5 coding sequence, the deletion of 32 nucleotides is expected to cause a frameshift mutation. This will change the downstream codons, produce a premature stop codon, and result in a shorter, truncated CCR5 protein with reduced or lost normal receptor function.

Methods

This study used Galaxy to analyze the CCR5 Δ32 mutation and compare it with the normal CCR5 gene. The wild-type (WT) CCR5 coding sequence (NM_000579.3) was imported into Galaxy and kept as the reference sequence throughout the analysis. A separate Galaxy history was created to organize the WT, documented mutant, and artificial mutant datasets.

The documented CCR5 Δ32 mutation was created by manually deleting 32 nucleotides (c.554–585) from a copy of the WT sequence. The WT and mutant CDS were then translated into protein sequences using the Transeq tool in Galaxy. The resulting proteins were compared based on protein length, reading frame, amino acid changes, and the presence of a premature stop codon.

A second experiment was also performed by creating an artificial one-nucleotide deletion (c.4delG). The artificial mutant was translated and compared with both the WT and documented mutant proteins to observe how different frameshift mutations affect the amino acid sequence and the predicted function of the CCR5 protein.

Part IV. Import the Normal Sequence into Galaxy

The WT CCR5 coding sequence (CDS) was uploaded into Galaxy and used as the reference sequence for the analysis. A separate Galaxy history was created to organize all files and results. The original WT sequence was kept unchanged and served as the control for both the documented mutation and the artificial mutation experiments.

Part V. Establish the Wild-Type Control

CDS length	1,059 bp
Predicted Protein length	352
Start codon	ATG
Stop codon	TGA
Reading frame	+1
First 10 amino acids	MDYQVSSPIY

Last 10 amino acids	TGEQEISVGL
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The translated protein matches the accepted RefSeq CCR5 protein (NP_000570.1). It has the expected 352 amino acids, begins with MDYQVSSPIY, and ends with TGEQEISVGL before the stop codon, indicating that the correct CDS and reading frame were used.

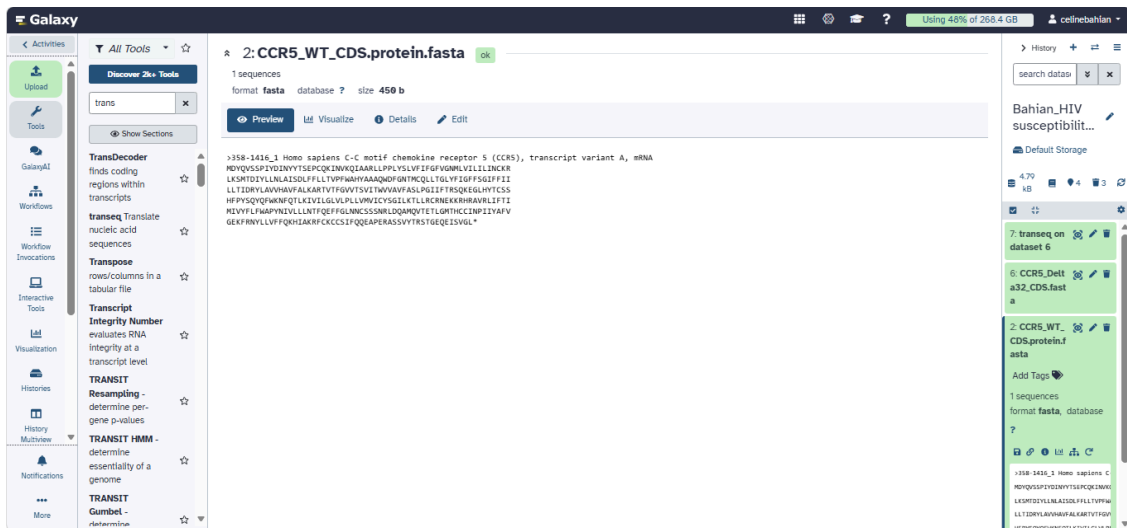


Figure 1. Predicted wild-type (WT) CCR5 protein sequence generated in Galaxy using the transeq translation tool. The translated amino acid sequence is shown in FASTA format and serves as the reference protein for comparison with the CCR5 $\Delta 32$ mutant.

Part VI. Formulate a Mutation Hypothesis

Mutation and exact nucleotide change	CCR5 $\Delta 32$: deletion of 32 nucleotides in the CCR5 coding sequence
Number of nucleotide(s) affected	32 nucleotides
Predicted mutation type	Frameshift deletion
Predicted effect on the reading frame	The reading frame shifts after the deletion, changing all downstream codons
Predicted effect on protein length	A premature stop codon is expected, producing a shorter (truncated) protein
Predicted effect on protein function	The truncated CCR5 protein is predicted to lose normal receptor function because it is incomplete

If the documented CCR5 $\Delta 32$ mutation is introduced into the WT CCR5 coding sequence, the deletion of 32 nucleotides is expected to cause a frameshift mutation. This will change the downstream codons, produce a premature stop codon, and result in a shorter, truncated CCR5 protein with reduced or lost normal receptor function

Part VII. Create the Mutant Sequence

Record the original nucleotide position(s)	C.554-585 CDS
original sequence	GTCAGTATCAATTCTGGAAGAATTTC AGACA
mutant sequence	Δ 554–585 (32-bp deletion)
number of bases inserted/deleted/substituted	32 nucleotides
mutation type	Deletion (32)

Part VIII. Translate the Mutant Sequence

Required Information	Result
Mutant CDS length	1,027bp
Mutant protein length	343 amino acids
Reading frame	Frame 1
First amino-acid difference	181
Premature stop codon	Present
Approximate number of amino acids affected	About 172 aa
Mutant protein FASTA filename	CCR5_MT_CDS.protein.fasta

The screenshot displays the Galaxy web interface. The main panel shows the output of the 'transeq' tool, which has translated a nucleotide sequence into a protein sequence. The output is displayed in FASTA format, starting with a header line: `>358-3432_1 Homo sapiens C-C motif chemokine receptor 5 (CCR5), transcript variant A, mRNA`. The protein sequence is shown in all caps: `MDYQVSSPVDYNYVSEPCDINWQZJARLLPPLVSLVDFGFGVGLVLLINCR`. The interface includes a sidebar with various tools and a history panel on the right.

Figure 2. Predicted amino acid sequence of the CCR5 Δ 32 mutant generated in Galaxy using the transeq translation tool. The translated protein sequence is shown in FASTA format and is used to evaluate the effect of the 32-base deletion on the resulting protein.

Part IX. Compare WT and Mutant Proteins

24. At what amino-acid position do the sequences first differ?	181
25. Is only one amino acid affected?	No
26. Are multiple downstream amino acids changed?	Yes
27. Was an amino acid deleted or inserted?	Deleted
28. Was a premature stop codon produced?	Yes
29. Did the reading frame change?	Yes
30. Did the protein length change?	Yes

The CCR5 $\Delta 32$ mutant was translated in Frame 1, but the 32-base deletion caused a frameshift starting around amino acid 181. This changed the remaining amino acid sequence and created a premature stop codon, producing a shorter 343-amino acid protein that is likely unable to function normally.

Part X. Explain the Molecular Consequence

CCR5 gene $\rightarrow$ c.554_585del32 mutation $\rightarrow$ 32-base deletion $\rightarrow$ frameshift (p.Ser185fs) $\rightarrow$ truncated CCR5 protein $\rightarrow$ loss of normal CCR5 receptor function $\rightarrow$ reduced HIV entry into CD4⁺ T cells $\rightarrow$ decreased susceptibility to HIV infection.

The CCR5 gene contains the instructions for making the CCR5 receptor, a protein found on the surface of immune cells such as CD4⁺ T cells. Normally, the DNA sequence is transcribed into mRNA, and the ribosome reads the mRNA three nucleotides at a time to produce the correct amino acid sequence. This results in a complete CCR5 protein that can be transported to the cell membrane and function as a chemokine receptor.

In the CCR5 $\Delta 32$ mutant, 32 nucleotides are deleted from the coding sequence. Because 32 is not a multiple of three, the deletion causes a frameshift mutation. After the deletion, the ribosome reads the codons incorrectly, so the amino acids produced after the mutation are completely different from the wild type. Our translated mutant sequence showed that a premature stop codon appeared, producing a 343-amino acid truncated protein instead of the normal full-length receptor.

The truncated CCR5 protein is missing important transmembrane regions that help the receptor fold correctly and attach to the cell membrane. As a result, the mutant receptor is unable to function normally. Since CCR5 is also used by most CCR5-tropic HIV-1 strains as a co-receptor for entering CD4⁺ T cells, the absence of a functional CCR5 receptor greatly reduces the virus's ability to infect the cell. Individuals who inherit two copies of the $\Delta 32$ mutation therefore have strong natural resistance to many HIV-1 infections.

Part XI. Second Experiment: Create Your Own Mutation

Artificial Mutation	One-nucleotide deletion
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Nucleotide Position	c.4delG
Original sequence	ATGGATTAT.....
Mutant Sequence	ATGATTAT.....
Number of bases affected	1 nucleotide base deleted
Predicted Mutation Type	Deletion
Predicted Effect in Reading Frame	The reading frameshift after nucleotide 4
Predicted Effect on Protein	Different amino acids are produced, followed by a premature stop codon, resulting in a truncated protein

For the second experiment, I created an artificial mutation by deleting one guanine (G) at position 4 of the CCR5 coding sequence (c.4delG). Since only one nucleotide is removed, the reading frame is expected to shift, changing all downstream codons. I predict that the translated protein will have a different amino acid sequence and will terminate early because of a premature stop codon, producing a shorter and likely nonfunctional CCR5 protein.

Part XII. Compare the Documented and Artificial Mutations

	Wild Type	Mutant Type	Artificial Type
CDS length,	1,059	1,027	1,058
protein length	353aa	343aa	353aa
mutation type	None	32-nt frameshift deletion	1-nt frameshift deletion
reading frame changed	No	Yes	Yes
premature stop codon appeared	No (normal stop only)	Yes	Yes
amino acids affected	Normal 353aa CCR5 sequence	Downstream amino acids altered after the deletion; truncated protein	Amino acids changed immediately after the deletion due to frameshift
expected functional consequence	Functional CCR5 receptor	Truncated, likely nonfunctional CCR5 receptor	Altered protein sequence, likely nonfunctional receptor

The comparison shows that the wild-type (WT) sequence produced a normal 353-amino acid CCR5 protein with no changes in the reading frame, indicating a functional receptor. In contrast, both the mutant type (MT) and the artificial type (AT) caused frameshift mutations that altered the reading frame and introduced a premature stop codon. These changes affected the downstream amino acid sequence, resulting in a truncated or altered protein that is likely unable to perform the normal function of the CCR5 receptor.

Part XIII. Interpretation Questions

32. Why does the exact location of a mutation matter?

The exact location matters because different parts of a gene have different functions. A mutation in the coding region can change the protein, while a mutation in another region may affect gene expression instead.

33. Why can deleting three nucleotides produce a different result from deleting one or two nucleotides?

Deleting three nucleotides removes one amino acid but keeps the reading frame the same. Deleting one or two nucleotides causes a frameshift and changes all the codons after the mutation.

34. Does every mutation change the amino-acid sequence? Explain.

No. Some mutations are synonymous, meaning the DNA changes but the amino acid stays the same, so the protein is unchanged.

35. Does every amino-acid substitution destroy protein function? Explain.

No. Some amino acid substitutions have little effect, especially if the new amino acid has similar properties. The effect depends on where the mutation occurs.

36. Why can a frameshift affect many amino acids even if only one nucleotide was deleted?

A frameshift changes how the ribosome reads every codon after the deletion. As a result, many amino acids become different until a stop codon is reached.

37. Why might a premature stop codon produce a nonfunctional protein?

A premature stop codon ends protein translation too early. This produces a shorter protein that may be missing important regions needed for its normal function.

38. Could a mutation affect protein function without greatly changing protein length?

Yes. A single amino acid change can affect the protein's shape or binding site, even if the overall protein length stays the same.

39. Could a mutation cause disease without changing the protein sequence? Give a possible molecular mechanism.

Yes. A mutation in a promoter or splice site can change how much protein is produced or how the mRNA is processed without changing the amino acid sequence.

40. What evidence from your analysis supports the proposed molecular mechanism of your disease?

Our Galaxy analysis showed that the CCR5 Δ 32 mutation deleted 32 nucleotides, caused a frameshift, and produced a truncated protein. The mutant protein sequence was different from the wild type, supporting the loss of normal CCR5 function.

41. Which conclusions are supported directly by your computational results, and which require evidence from published experimental studies?

The computational results directly showed the DNA deletion, frameshift, and shortened protein. The conclusion that this mutation reduces HIV entry and provides resistance is supported by published experimental studies

Part XV. GitHub Commit History

https://github.com/celinebahian/HIV_susceptibility-Gene-Mutation-Lab

Galaxy link:

<https://usegalaxy.org/u/celinebahian/h/bahian-hiv-susceptibility-gene-mutation-lab>

Part XVI. Limitations

This study was limited to a computational analysis using Galaxy and did not include laboratory experiments to verify the predicted effects of the CCR5 Δ 32 mutation. The results were based on sequence translation and comparison of the wild-type, documented mutant, and artificial mutant proteins, so the actual protein activity and HIV resistance were not directly measured. In addition, only one documented mutation and one artificial mutation were analyzed, which means other CCR5 variants and genetic factors involved in HIV susceptibility were not included.

Part XVII. Conclusion

The results of this study showed that the CCR5 Δ 32 mutation causes a 32-base deletion that changes the reading frame and produces a shorter, truncated CCR5 protein. Compared with the wild-type sequence, both the documented mutant and the artificial mutation disrupted the normal amino acid sequence, demonstrating how frameshift mutations can greatly affect protein structure and function. Overall, the Galaxy analysis helped explain the relationship between DNA mutations, protein changes, and the molecular basis of reduced CCR5 function associated with HIV resistance.

XV. REFERENCES

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